

Kambala Church of England Girls' School

Trial Higher School Certificate Examination , 2000

Year 12

August, 2000

MATHEMATICS

3/4 UNIT

Time Allowed : 2 hours (plus 5 minutes reading time)



LORETO KIRRIBILLI
85 CARABELLA ST
KIRRIBILLI 2061 (16)

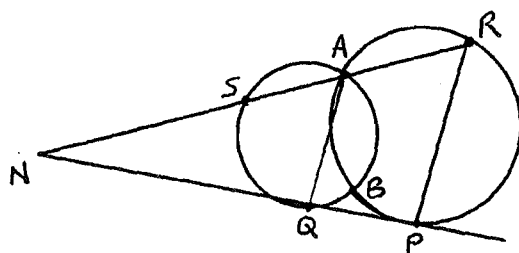
Instructions :

- ALL questions may be attempted.
- ALL questions are of equal value.
- All necessary working should be shown in every question.
- Full marks may not be awarded for careless or badly arranged work.
- Approved scientific calculators and drawing templates may be used.
- A Table of Standard Integrals is contained at the end of the examination paper.
- Start each question in a NEW BOOK.
- This is a Trial Paper only, and does NOT necessarily reflect either the content or format of the final HSC Examination.

Question 1 :

(a) Solve $\frac{x^2-1}{x} > 0$.

- (b) Two circles intersect at A and B and a common tangent touches them at P and Q.
PR // QA.
RA is produced to cut the other circle at S and the tangent at N.
Prove that PRSQ is a cyclic quadrilateral.



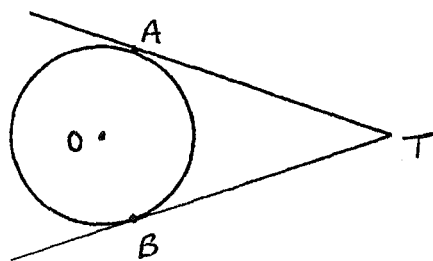
(c) Find $\int_0^{\frac{2}{3}} \frac{dx}{4+9x^2}$.

- (d) Find the equation of the two lines through the point (5,3) which make acute angles of $\frac{\pi}{4}$ with the line $2x - y + 2 = 0$.

Question 2 : (Start a NEW BOOK)

- (a) TA and TB are tangents to the circle drawn below, with centre O.

Prove that : $\frac{\angle OAB}{\angle ATB} = \frac{1}{2}$.



- (b) Find the general solution of the equation $\sin 2\theta = \sin \theta$, θ measured in radians.
- (c) Prove by the Principle of Mathematical Induction that $3^{3^n} + 2^{n+2}$ is a multiple of 5 for all positive integers n.

Question 3 : (Start a NEW BOOK)

- (a) The point $P(1,6)$ divides the interval AB in the ratio $m:n$.
If $A = (7,0)$ and $B = (3,4)$, find the value of the ratio $m:n$.

- (b) Find $\frac{d}{dx}(x \sin^{-1} 2x + \frac{1}{2}\sqrt{1-4x^2})$.

Hence evaluate $\int_0^{\frac{1}{2}} \sin^{-1} 2x \, dx$.

- (c) (i) If $t = \tan \frac{\theta}{2}$, find $\cos \theta$ and $\sin \theta$ in terms of t .
(ii) Hence solve the equation $3 \sin \theta + 4 \cos \theta = 5$ for values of θ in the range $0^\circ \leq \theta \leq 360^\circ$.

Question 4 : (Start a NEW BOOK)

- (a) Evaluate the following definite integrals using the substitutions given :

(i) $\int_0^3 \frac{x}{\sqrt{4-x}} \, dx$ substitute $u = 4 - x$.

(ii) $\int_0^2 \frac{dx}{(4+x^2)^2}$ substitute $x = 2 \tan \theta$.

- (b) The polynomial $P(x) = ax^3 + bx^2 - 8x + 3$ has a factor of $(x-1)$ and leaves a remainder of 15 when divided by $(x+2)$.

- (i) Find the values of a and b .
(ii) Hence, factorise $P(x)$ fully and sketch the curve.
(iii) Determine the set of values of x for which $P(x) > 0$.

Question 5 : (Start a NEW BOOK)

- (a) $P(2ap, ap^2)$ is any point on the parabola $x^2 = 4ay$. S is the focus $(0, a)$. The tangent to the parabola at P meets the Y -axis in M . The perpendicular to the tangent PM from S meets PM in N . Find :
- (i) the co-ordinates of M and N .
 - (ii) the co-ordinates of the midpoint K of MN .
 - (iii) the equation of the locus of K as P varies.
- (b) A circular oil slick lies on the surface of a body of calm water. If its area is increasing at the rate of $1500 \text{ m}^2/\text{h}$, at what rate is its circumference increasing when the radius of the slick is 1250 m .

Question 6 : (Start a NEW BOOK)

A stone is thrown horizontally with a velocity of 20 m/s from the top of a tower 100 m high. Assuming no air resistance, and that the acceleration due to gravity, $g \approx 10 \text{ m/s}^2$;

- (i) express x and y in terms of t .
- (ii) find the equation of the trajectory.
- (iii) find how long the stone takes to reach the ground .
- (iv) find how far from the foot of the tower the stone strike the ground .
- (v) find the velocity and direction of the stone on impact with the ground .

Question 7 : (Start a NEW Page)

- (a) Define Simple Harmonic Motion.
- (b) A particle moves from the origin, O with velocity $(2p)$ m/s, and is subject to a retardation of its motion equal to q times its distance x from the origin ($q > 0$).
(Note : retardation means negative acceleration)
- (i) Show that the distance it travels, before coming to rest is $\frac{2p}{\sqrt{q}}$ metres.
- (ii) Find the time when the particle first comes to rest.
- (iii) Find where the particle is after $\frac{\pi}{4\sqrt{q}}$ seconds.

END OF EXAM

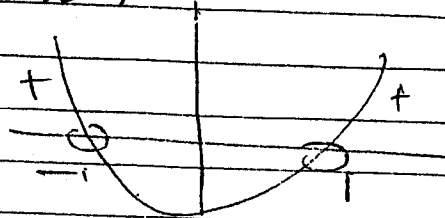
KAMBALA

Q1 (a) $\frac{x^2-1}{x} > 0$

$x \neq 0: x^2-1 > 0$

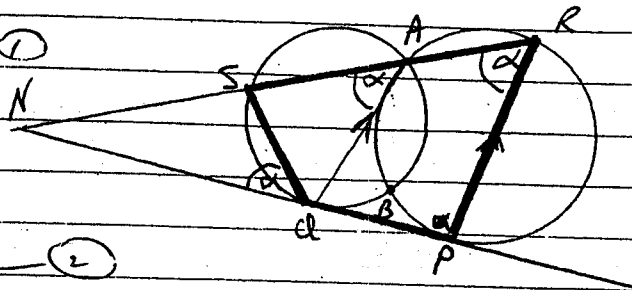
$(x-1)(x+1) > 0$

$x > 1, x < -1$ — 2



(b) $\angle SAQ = \angle ARP = \alpha$ — (1)

(corr. $\angle$ s = in
// lines PK, AQ)



$\angle SAQ = \angle SQN = \alpha$ — (2)

(angle between a tangent and a chord = angle in the alternate segment)

$\therefore \angle SQN = \angle SRP$

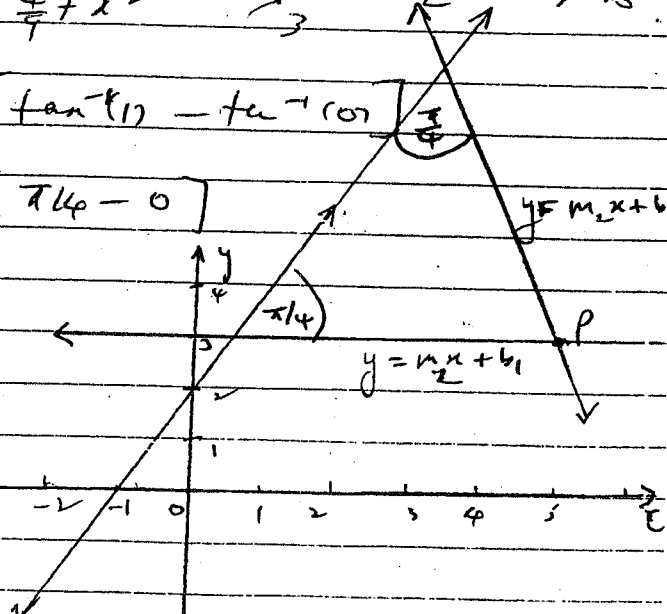
$\therefore PQRS$ is a cyclic quad ($\text{ext } \angle = \text{inter. opp } \angle$)

(c) $\int_0^{2/3} \frac{dx}{4+9x^2} = \frac{1}{9} \int_0^{2/3} \frac{dx}{\frac{4}{9}+x^2} = \frac{1}{9} \cdot \frac{3}{2} \left[\tan^{-1}\left(\frac{3x}{2}\right) \right]_0^{2/3}$

$= \frac{1}{6} \left[\tan^{-1}(1) - \tan^{-1}(0) \right]$

$= \frac{1}{6} \left[\frac{\pi}{4} - 0 \right]$

$= \frac{\pi}{24}$



(d) $2x - y + 2 = 0$

x	0	-1
y	2	0

$m_1 = 2$

$\tan \theta = \left| \frac{m_1 - m_2}{1 + m_1 m_2} \right| \Rightarrow$

Q1cd

$$\tan \theta = \left| \frac{m_1 - m_2}{1 + m_1 m_2} \right|$$

$$m_2 = \frac{1}{3}, (5, 3)$$

$$y - y_1 = m(x - x_1)$$

$$y - 3 = \frac{1}{3}(x - 5)$$

$$3y - 9 = x - 5$$

$$x - 3y + 4 = 0$$

$$\tan \pi/4 = \left| \frac{2 - m_2}{1 + 2m_2} \right|$$

$$m_2 = -3, (5, 3)$$

4

$$1 = \left| \frac{2 - m_2}{1 + 2m_2} \right|$$

$$1 + 2m_2 = \pm (2 - m_2)$$

$$y - 3 = -3(x - 5)$$

$$y - 3 = -3x + 15$$

$$3x + y - 18 = 0$$

$$1 + 2m_2 = 2 - m_2$$

$$1 + 2m_2 = -2 + m_2$$

$$3m_2 = 1$$

$$m_2 = -3$$

$$m_2 = 1/3$$

Q1 a) Let $\angle OAB = \alpha$

and $\angle ATB = 2\beta$.

and $\angle BAT = \theta$

$\triangle OAT \equiv \triangle OBT$ (RHS)

$\therefore \angle ATO = \angle BTO = \beta$

(Comp. L's of $\triangle OAT$ and $\triangle OBT$)

$\triangle AOT \equiv \triangle BOT$ (SAS)

$\therefore \angle AOT = \angle BOT = 90^\circ$ (= supp. L's of a str. line)

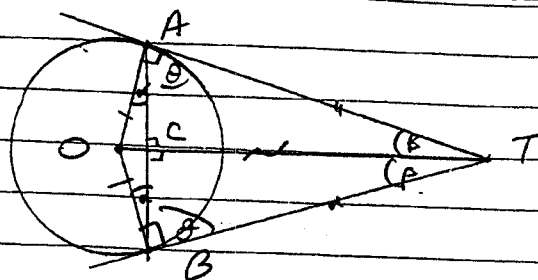
In $\triangle OAT$, $\alpha + \theta = 90^\circ$ (Comp. L's of a str. line)

In $\triangle OBT$, $\beta + \theta = 90^\circ$ (Comp. L's of a str. line)

$\therefore \alpha = \beta$

$\therefore \angle ATB = 2\beta = 2\alpha = 2 \times \angle OAB$

$\therefore \frac{\angle OAB}{\angle ATB} = \frac{1}{2}$



4

b) $\sin 2\theta = \cos \theta$

$2 \sin \theta \cos \theta = \cos \theta$

$\sin \theta (2 \cos \theta - 1) = 0$

$\therefore \sin \theta = 0$ or $\cos \theta = \frac{1}{2}$

$\theta = n\pi$ or $\theta = 2n\pi \pm \frac{\pi}{3}$

c) $3^{3n} + 2^{n+2}$ is a multiple of 5

S1 $n=1$ $3^{3n} + 2^{n+2} = 3^3 + 2^3 = 27 + 8 = 35 = 5 \times 7$

$\therefore$ True for $n=1$

S2 $n=k$: $3^{3k} + 2^{k+2} = 5p$

$\therefore 3^{3k} = 5p - 2^{k+2}$ — (A)

S3: $n=k+1$ $3^{3(k+1)} + 2^{(k+1)+2} = 5q$ — (B)

(B) LHS $= 3^{3k+3} + 2^{k+3}$

$= 27(3^{3k}) + 2(2^{k+2})$

$= 27(5p - 2^{k+2}) + 2(2^{k+2})$

$= 5(27p) - 2^{k+2}(27 - 2)$

$$\begin{aligned}
 Q2(c) \quad LHS &= 5(27p) - 2^{k+2}(25) \\
 &= 5[27p - 5(2^{k+2})] \\
 &= 5Q \\
 &= 2(15)
 \end{aligned}$$

$\therefore$ By Induction true for all the natural n .



Q3 c),

A (7, 0)

B (3, 4)

+ m : n

$$P = \left(\frac{+3m+7n}{+m+n}, \frac{4m+0n}{m+n} \right)$$

$$P = (1, 6)$$

$$\therefore \frac{3m+7n}{m+n} = 1$$

$$3m+7n = m+n$$

$$2m = -6n$$

$$m = -3n$$

$$\therefore \frac{m}{n} = -3$$

$$m:n = -3:1$$

$$(P(4n=1))$$

(b)

$$\frac{d}{dx} \left(\frac{x \cdot \sin^{-1} 2x}{\sqrt{1-4x^2}} + \frac{1}{2} \sqrt{1-4x^2} \right)$$

$$= \sin^{-1} 2x \cdot 1 + x \cdot \frac{2}{\sqrt{1-4x^2}} + \frac{1}{2} \cdot \left(\frac{1}{2} \right) \cdot (-4x^2) \cdot \frac{1}{\sqrt{1-4x^2}}$$

$$= \sin^{-1}(2x) + \frac{2x}{\sqrt{1-4x^2}} - \frac{2x}{\sqrt{1-4x^2}}$$

$$= \sin^{-1}(2x)$$

4

$$= \int_0^{\frac{1}{2}} \sin^{-1}(2u) du = \left[u \sin^{-1}(2u) + \frac{1}{2} \sqrt{1-4u^2} \right]_0^{\frac{1}{2}}$$

$$= \left[\frac{1}{2} \sin^{-1}(1) + 0 \right] - \left[0 + \frac{1}{2} \right]$$

$$= \frac{1}{2} \left(\frac{\pi}{2} - 1 \right)$$

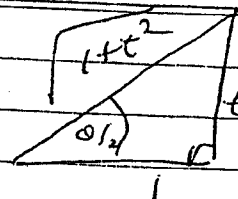
$$= \frac{1}{2} \left(\frac{\pi}{2} - 1 \right)$$

Q3 (c) (i) $t = \tan \theta$

$$\cos \theta = \cos \frac{\theta}{2} - \sin \frac{\theta}{2}$$

$$= \frac{1}{1+t^2} - \frac{t}{1+t^2}$$

$$\cos \theta = \frac{1-t}{1+t}$$



$$\tan \frac{\theta}{2} = \frac{t}{1}$$

$$\sin \frac{\theta}{2} = \frac{t}{\sqrt{1+t^2}}$$

$$\cos \frac{\theta}{2} = \frac{1}{\sqrt{1+t^2}}$$

$$\sin \theta = 2 \sin \frac{\theta}{2} \cos \frac{\theta}{2}$$

$$= 2 \cdot \frac{t}{\sqrt{1+t^2}} \cdot \frac{1}{\sqrt{1+t^2}}$$

$$\sin \theta = \frac{2t}{1+t^2}$$

2

(ii) $3 \sin \theta + 4 \cos \theta = 5$

$$3 \left(\frac{2t}{1+t^2} \right) + 4 \left(\frac{1-t^2}{1+t^2} \right) = 5$$

$$6t + 4 - 4t^2 = 5 + 5t^2$$

$$9t^2 - 6t + 1 = 0$$

$$(3t-1)^2 = 0$$

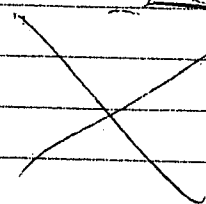
$$t = \frac{1}{3}$$

$$\tan \frac{\theta}{2} = \frac{1}{3}$$

$$\theta = 18^\circ 26'$$

$$\theta = 36^\circ 52'$$

3



Q4

cal 'i'

$$\int_0^3 \frac{x}{\sqrt{4-x}} dx = I$$

$$u = 4-x$$
$$du = -dx$$

$$\begin{array}{ccc} x & 0 & 3 \\ u & 4 & 1 \end{array}$$

$$I = \int_4^1 \frac{4-u}{\sqrt{u}} - du$$

$$= + \int_1^4 \frac{4-u}{\sqrt{u}} du$$

$$= + \int_1^4 (4u^{-1/2} - u^{1/2}) du$$

$$= \left[\frac{4u^{1/2}}{1/2} - \frac{u^{3/2}}{3/2} \right]_1^4$$

$$= \left[8\sqrt{u} - \frac{2}{3}\sqrt{u^3} \right]_1^4$$

$$= \left(8\sqrt{4} - \frac{2}{3}\sqrt{4^3} \right) - \left(8\sqrt{1} - \frac{2}{3}\sqrt{1^3} \right)$$

$$= \left(16 - \frac{16}{3} \right) - \left(8 - \frac{2}{3} \right)$$

$$= 8 - \frac{14}{3}$$

$$= \frac{10}{3}$$

$$\boxed{x = 3 \frac{1}{3}}$$

Q. 1. (ii) $I = \int_0^2 \frac{du}{(4+u^2)^{3/2}}$

$u = 2 \tan \theta$
 $du = 2 \sec^2 \theta d\theta$

$\begin{matrix} u & 0 & 2 \\ \theta & 0 & \pi/4 \end{matrix}$

$I = \int_0^{\pi/4} \frac{2 \sec^2 \theta d\theta}{4 \sqrt{4} \sec^3 \theta}$

$= \frac{1}{4} \int_0^{\pi/4} \cos \theta d\theta$

$= \frac{1}{4} \left[\sin \theta \right]_0^{\pi/4}$

$= \frac{1}{4} (\sin \pi/4 - \sin 0)$

$= \frac{1}{4} (\frac{1}{\sqrt{2}} - 0)$

$= \frac{1}{4\sqrt{2}}$

3

$I = \frac{1}{4\sqrt{2}}$

Q4 (b) $P(x) = x^3 - 4x^2 + 3x + 2 = 0$

$$\begin{aligned} \alpha + \beta + \gamma &= -4/a = 4 \\ \alpha\beta + \beta\gamma + \gamma\alpha &= c/a = 3 \\ \alpha\beta\gamma &= -d/a = -2 \end{aligned} \quad \left. \vphantom{\begin{aligned} \alpha + \beta + \gamma &= -4/a = 4 \\ \alpha\beta + \beta\gamma + \gamma\alpha &= c/a = 3 \\ \alpha\beta\gamma &= -d/a = -2 \end{aligned}} \right\} 3$$

(i) $\frac{1}{\alpha} + \frac{1}{\beta} + \frac{1}{\gamma} = \frac{\beta\gamma + \alpha\gamma + \alpha\beta}{\alpha\beta\gamma} = \frac{3}{-2} = -\frac{3}{2}$

(ii) $\alpha^2 + \beta^2 + \gamma^2 = (\alpha + \beta + \gamma)^2 - 2(\alpha\beta + \beta\gamma + \gamma\alpha)$
 $= 4^2 - 2(3)$
 $= 16 - 6$
 $= 10$

Q4 (b) (i) $P(x) = ax^3 + bx^2 - 8x + 3$

$(x-1)$ is a factor of $P(x)$, $\therefore P(1) = 0$

$$P(1) = a(1)^3 + b(1)^2 - 8(1) + 3$$

$$0 = a + b - 8 + 3$$

$$\boxed{a + b = 5} \quad \text{--- (1)}$$

remainder = 15 when $\div$ by $(x+2)$, $\therefore P(-2) = 15$

$$15 = a(-2)^3 + b(-2)^2 - 8(-2) + 3$$

$$0/15 = -8a + 4b + 16 + 3$$

$$8a - 4b = 4$$

$$\boxed{2a - b = 1} \quad \text{--- (2)}$$

① + ②:

$$3a = 6$$

$$\boxed{a = 2}$$

Sub $a = \frac{1}{2}$ into (2):

$$2(2) - b = 1$$

$$4 - b = 1$$

$$\boxed{b = 3}$$

(ii) $P(x) = 2x^3 + 3x^2 - 8x + 3 = (x-1)(2x^2 + 5x - 3)$

$$\begin{array}{r} 2x^2 + 5x - 3 \\ x-1 \overline{) 2x^3 + 3x^2 - 8x + 3} \\ \underline{2x^3 - 2x^2} \\ 5x^2 - 8x \\ \underline{5x^2 - 5x} \\ -3x + 3 \\ \underline{-3x + 3} \\ 0 \end{array}$$

$$\begin{array}{|l} P = -6 \\ S = 5 \\ F = 6, -1 \end{array}$$

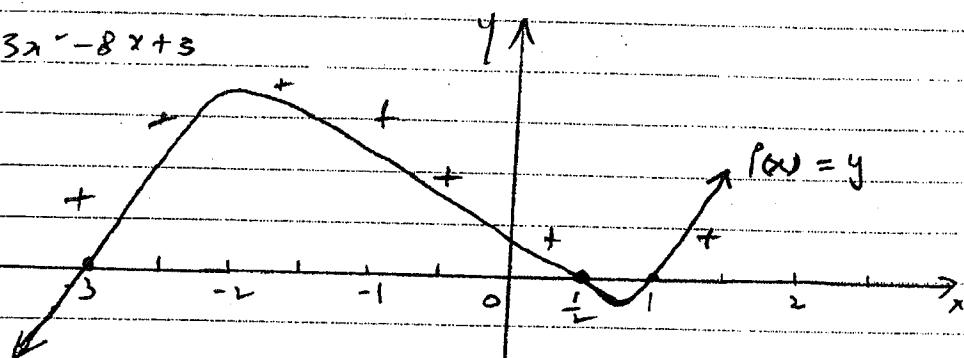
Q4 (b), (ii): $P(x) = 2x^3 + 3x^2 - 8x + 3$
 $= (x-1)(2x^2 + 5x - 3)$

$$\begin{aligned} 2x^2 + 5x - 3 &= 2x^2 + 6x - 1x - 3 \\ &= 2x(x+3) - 1(x+3) \\ &= (x+3)(2x-1) \end{aligned}$$

$$\therefore P(x) = (x+3)(2x-1)(x-1)$$

2

$$P(x) = 2x^3 + 3x^2 - 8x + 3$$



(iii) $P(x) > 0$, from the graph occurs when

$$-3 < x < \frac{1}{2} \text{ \& } x > 1$$

(15) Target ctf 1

1) $y - px + ap^2 = 0$ — (1)

At M, $x = 0$

$$y = -ap^2$$

2

$M = (0, -ap^2)$

$\underline{SN} : m = -\frac{1}{p}, S(0, a)$

$$y - a = -\frac{1}{p}(x - 0)$$

$$py - pa = -x$$

$$x + py = pa \quad \text{--- (2)}$$

To find N, solve (1) & (2) simultaneously

$$(1) \Rightarrow y - px = -ap^2$$

$$y = px - ap^2 \quad \text{--- (3)}$$

Sub into (2)

$$x + p(px - ap^2) = pa$$

$$x + p^2x - ap^3 = pa$$

$$x(1 + p^2) = pa + ap^3$$

$$x = \frac{pa(1 + p^2)}{1 + p^2}$$

$$x = pa$$

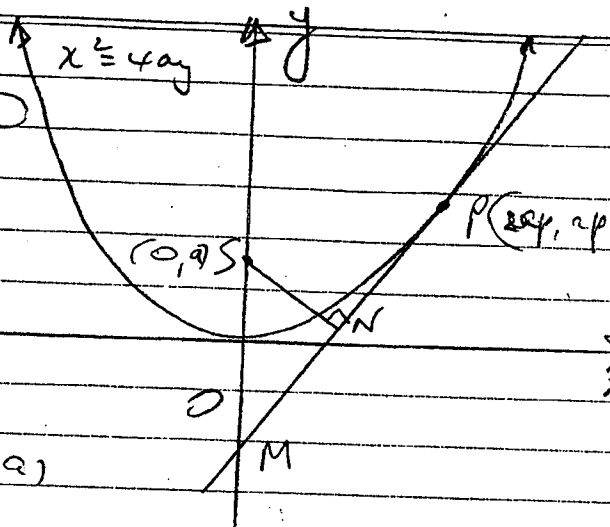
$$y = p(pa) - ap^2 = p^2a - ap^2 = ap^2$$

$$y = ap^2$$

$$N = (pa, ap^2)$$

$$M = (0, -ap^2)$$

$$MN = \sqrt{(pa - 0)^2 + (ap^2 + ap^2)^2} = \sqrt{p^2a^2 + 4a^2p^4} = pa\sqrt{1 + 4p^2}$$



$$\textcircled{05} \text{ (i)} \quad x + p^2 x - ap^3 = ap$$

$$(1+p^2)x = ap + ap^3$$

$$(1+p^2)x = ap(1+p^2)$$

$$x = ap$$

Sub $x = ap$ into $\textcircled{3}$

$$4 \quad y = p(ap) - ap^2$$

$$\therefore y = 0$$

$$\therefore \boxed{M = (ap, 0)}$$

$$\textcircled{\text{ii}} \quad K = \text{mid pt of } MN$$

$$M = (ap, 0)$$

$$N = (0, -ap^2)$$

$$\therefore K = \left(\frac{ap+0}{2}, \frac{0+(-ap^2)}{2} \right)$$

$$\boxed{K = \left(\frac{ap}{2}, -\frac{ap^2}{2} \right)}$$

$$\textcircled{\text{iii}} \quad x = ap \quad \textcircled{1} \Rightarrow \quad \textcircled{p = \frac{2x}{a}} \quad \textcircled{3}$$

$$y = -\frac{ap^2}{2} \quad \textcircled{2}$$

Sub $\textcircled{3}$ into $\textcircled{2}$:

$$y = -\frac{a}{2} \left(\frac{2x}{a} \right)^2$$

$$= -\frac{a}{2} \left(\frac{4x^2}{a^2} \right)$$

$$y = -\frac{2x^2}{a}$$

$$\therefore \boxed{x^2 = -\frac{1}{2}ay}$$

~~$$\frac{\cos A - \cos(A+B)}{2 \sin B} = \sin(A+B)$$~~

~~$$\text{LHS} = \frac{\cos A - \cos A \cos 2B + \sin A \sin 2B}{2 \sin B}$$~~

~~$$= \frac{\cos A(1 - \cos 2B) + \sin A(2 \sin B \cos B)}{2 \sin B}$$~~

~~$$= \frac{\cos A(2 \sin^2 B) + \sin A(2 \sin B \cos B)}{2 \sin B}$$~~

~~$$= \frac{2 \sin B (\sin B \cos A + \cos B \sin A)}{2 \sin B}$$~~

~~$$= \sin(B+A)$$~~

~~$$= \sin(A+B)$$~~

~~$$= \text{RHS}$$~~

5 (h)

$$\frac{dA}{dt} = 1500 \text{ m}^2/\text{h}$$

$$r = 1250 \text{ m}$$

$$C = 2\pi r$$

$$\frac{dC}{dt} = 2\pi \times \frac{dr}{dt}$$

$$= 2\pi \times \frac{3}{5}$$

$$\frac{dC}{dt} = \frac{6}{5} \text{ m/h}$$

$$A = \pi r^2$$

$$\frac{dA}{dt} = 2\pi r \frac{dr}{dt}$$

$$1500 = 2\pi \times 1250 \times \frac{dr}{dt}$$

$$\frac{dr}{dt} = \frac{1500}{2500\pi}$$

$$\frac{dr}{dt} = \frac{3}{5\pi} \text{ m/h}$$

Q1

$$g = 10 \text{ m/s}^2$$

$$\text{ii) } \ddot{x} = 0$$

$$\dot{x} = V \cos \theta$$

$$\dot{x} = 20 \times \cos 0$$

$$\dot{x} = 20$$

$$x = 20t$$

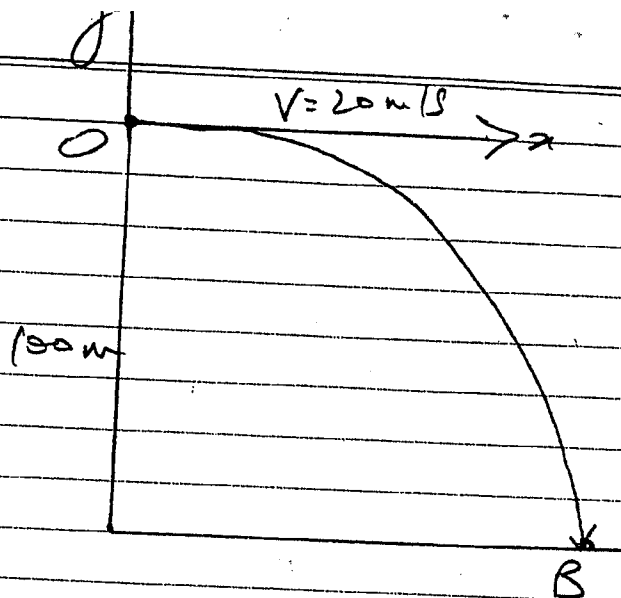
$$\ddot{y} = -10$$

$$\dot{y} = -10t + V \sin \theta$$

$$\dot{y} = -10t + 20 \times 0$$

$$\dot{y} = -10t$$

$$y = -5t^2$$



2

(iii) From (i)

$$x = 20t \Rightarrow t = \frac{x}{20}$$

2 (ii)

$$y = -5t^2$$

$$\therefore y = -5 \left(\frac{x}{20} \right)^2$$

$$= \frac{-5 \cdot x^2}{400}$$

2

$$y = -\frac{x^2}{80}$$

$$x^2 = -80y$$

(iii)

$$y = -100$$

$$-5t^2 = -100$$

$$t^2 = 20$$

$$t = \sqrt{20} \text{ sec}$$

2

$$(iv) t = \sqrt{20} \quad x = 20t$$

$$R = 20\sqrt{20} \text{ m}$$

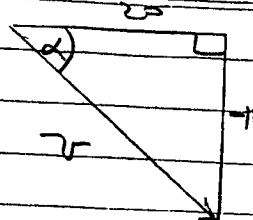
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Q6

$$r = \sqrt{20}$$

$$\dot{x} = 20$$

$$\dot{y} = -10(\sqrt{20})$$



$$v^2 = \dot{x}^2 + \dot{y}^2$$

$$v^2 = (20)^2 + (-10\sqrt{20})^2$$

$$v^2 = 400 + 2000$$

$$v^2 = 2400$$

$$v = \sqrt{2400}$$

$$v = 20\sqrt{6} \text{ m/s}$$

$$\tan \alpha = \frac{\dot{y}}{\dot{x}}$$

$$= \frac{-10\sqrt{20}}{20}$$

$$\tan \alpha = -\frac{\sqrt{20}}{2}$$

$$\alpha = -28.5^\circ$$

$$\tan \alpha = -\sqrt{r}$$

$$\alpha = 180 - 65.5^\circ$$

$$\alpha = 114.5^\circ$$

Q7

(a) A particle is travelling in a straight line and

(i) it is travelling in a straight line and

(ii) if it experiences an acceleration of the form $\ddot{x} = -k^2 x$

hence when $t = 0$, $x = 0$, $v = 2p$ m/s

$$\ddot{x} = -g x$$

$$\frac{d}{dt} \left(\frac{1}{2} v^2 \right) = -g x$$

$$\therefore \frac{1}{2} v^2 = -\frac{g x^2}{2} + c$$

$$\left. \begin{array}{l} x=0 \\ v=2p \end{array} \right\} \frac{1}{2} (2p)^2 = -\frac{g(0)^2}{2} + c$$

$$\therefore c = 2p^2$$

$$\therefore \frac{1}{2} v^2 = -\frac{g x^2}{2} + 2p^2$$

$$\therefore v^2 = 4p^2 - g x^2$$

when it comes to rest, $v = 0$.

$$\therefore 0 = 4p^2 - g x^2$$

$$\therefore x^2 = \frac{4p^2}{g}$$

$$\therefore x = \frac{2p}{\sqrt{g}} \text{ m}$$

Q7 (b) (ii) $x = a \cos(\omega t + \phi)$

when $t = 0$, $x = 0$ $\therefore$

$$\therefore 0 = a \cos \phi$$

$$\therefore \phi = \pi/2$$

$$\therefore x = a \cos(\omega t + \pi/2)$$

$$= a \left[\cos \omega t \cdot \cos \frac{\pi}{2} - \sin \omega t \cdot \sin \frac{\pi}{2} \right]$$

$$x = -a \sin \omega t$$

At $t = 0$, $x = \frac{2p}{\sqrt{g}} = a$

$$\therefore \sin \omega t = -1$$

when $\omega^2 = g$

$$\therefore \omega = \sqrt{g}$$

$$\therefore \sin(\sqrt{g} t) = -1$$

$$\therefore \sqrt{g} t = \frac{3\pi}{2}$$

$$\therefore t = \frac{3\pi}{2\sqrt{g}} \text{ sec}$$

(iii) $t = \frac{\pi}{2\sqrt{g}} = \frac{\pi}{4\sqrt{g}}$

$$x = -\frac{2p}{\sqrt{g}} \sin(\sqrt{g} t)$$

$$= -\frac{2p}{\sqrt{g}} \left(\sin \sqrt{g} \cdot \frac{\pi}{4\sqrt{g}} \right)$$

$$= -\frac{2p}{\sqrt{g}} \sin \pi/4 = -\frac{2p}{\sqrt{g}} \cdot \frac{1}{\sqrt{2}}$$

$$|x| = \frac{\sqrt{2}}{\sqrt{g}} p \text{ m}$$